

Pixaflip Platform

Project Requirements Document (PRD)

Date: April 14, 2026

Status: Version 1.0

1. PROJECT OVERVIEW

Pixaflip is a specialized educational management and workshop delivery platform. It is designed to bridge the gap between technical mentors, college administrations, and students, while automating the heavy administrative burden of academic accreditations like **NAAC (National Assessment and Accreditation Council)** and **NBA (National Board of Accreditation)**.

2. USER HIERARCHY & AUTHENTICATION FLOW

The platform follows a strict, non-public onboarding process to ensure only verified participants of active workshops can access the system.

Role	Created By	Primary Responsibility
Pixaflip Admin	System Root	Global oversight, analytics, and college onboarding.
College Admin	Pixaflip Admin	Managing faculty, oversight of student progress, and accreditation.
Teacher	College/Pixaflip Admin	Class-level management, engagement tracking, and grading.
Student	Teacher	Curriculum engagement, testing, and project submission.

Onboarding Process

- **No Public Sign-up:** No open registration; restricted access.
- **Verification:** Users added manually or via bulk invite by the tier above them.
- **College Selection:** Students are mapped to specific institutions (e.g., Keystone School of Engineering) during profile setup.

3. ROLE-BASED FEATURE SPECIFICATIONS

A. Pixaflip Admin (Global Control)

- **Global Dashboard:** Access to analytical data from all participating colleges.
- **College Management:** Onboard new colleges and set workshop schedules.
- **Full Oversight:** View all discussion threads, student submissions, and teacher activity.

B. College Admin (Oversight & Reporting)

- **Teacher Management:** Create profiles for professors (e.g., HOD creating a profile for a subject teacher).
- **Reporting Tools:** Generate reports on student engagement and skills benchmarks.
- **Feedback Management:** Review student feedback surveys for workshop improvements.
- **Support Channel:** Direct communication with teachers and Pixaflip support.

Note: Admins cannot edit the core curriculum or upload new workshop content.

C. Teacher (Instruction & Monitoring)

- **Class Management:** Manage specific divisions (e.g., 10th-A, 60 students).
- **Engagement Tracking:** Compare active website presence vs. physical attendance.

- **Activity Checklist:** Track document opens, video views, or test uploads.
- **Grading & Feedback:** Grade tests and provide private comments on student projects.
- **Templates:** Access to ready-made templates for announcements and feedback forms.

D. Student (Learning & Participation)

- **Personal Dashboard:** Overview of upcoming tasks, pending assignments, and unopened content.
- **Learning Hub:** Access to curriculum, video streaming, and technical resources.
- **Submissions:** Interface for taking tests and uploading "Mini Projects."
- **News & Announcements:** Real-time updates from teachers or Pixafliip.

4. SYSTEM-WIDE FEATURES

Attendance Module

Student Side: Self-mark attendance for workshops via the platform.

Verification: Teachers and College Admins cross-check logs with authority to override and mark "Absent" if required.

Collaboration & Gamification

- **Discussion Forums:** Thread-based forums for cross-role communication.
- **Gamification:** Badges and milestones for module completion and high scores.
- **Coding Sandbox:** Integrated browser-based environment to write, run, and preview code.

5. AI-POWERED ACCREDITATION ENGINE (NAAC & NBA)

This module automates official documentation required for college accreditations.

Data Inputs: Live Photos (workshop candid), Workshop Conclusions (raw notes), Assessment Data (test results/scores).

AI Processing: Measures "Impact" and "Effectiveness" from metadata and results.

Outputs:

- Automatically generated conclusion reports.
- Skill assessment charts.
- Formatted documentation ready for NAAC and NBA submission.

6. TECHNICAL WORKFLOW

1. **Deployment:** Pixaflip Admin sets up the college profile.
2. **Onboarding:** Cascading creation (College Admin → Teacher → Student).
3. **Delivery:** Workshop content delivery; attendance tracking; coding sandbox usage.
4. **Assessment:** Student testing; teacher grading via checklist.
5. **Accreditation:** Raw data processed by AI to generate final college reports.